AE 6210: ADVANCED DYNAMICS I

SPRING 2025 HOMEWORK #6

ANIKA TASNEEM

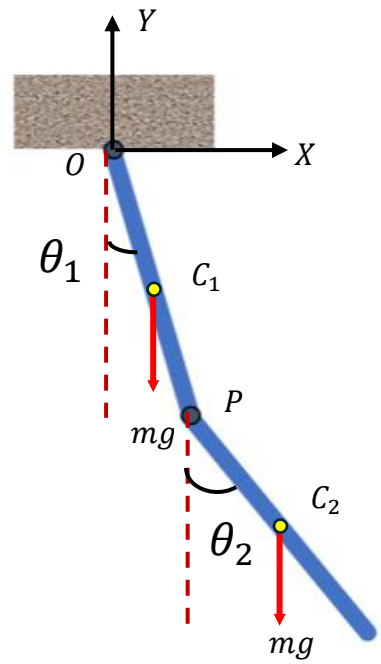


Fig: A Double Compound Pendulum

Problem Statement

We are given a double compound pendulum moving in the X-Y plane. The rods are of equal lengths (L) and have uniformly distributed mass (m). θ_1 and θ_2 are the angles the two rods make with the vertical. We have to derive the equations of motion of this system using:

a. **Lagrangian Mechanics (Hamilton's principle and Lagrange's equations)**
b. **Newtonian Mechanics (Newton's laws and Euler-Newton equations)**

Solution:

In the case of pendulum problems, it is generally in our best interest to do all the analysis and computation using the Cartesian coordinate system. Due to the geometry of the pendulum system and the fact that each mass is a rectangular rod, we can use the formula for the Moment of Inertia about their respective CoM to be $I^C = \frac{1}{12} mL^2$

Deriving the Equations of Motion using the Lagrangian Mechanics:

To use the **Method of Lagrangian Mechanics** for deriving the EoMs, we follow the steps which are given below:

Step #01: Defining the Generalized Coordinates

We choose the generalized coordinates to be $\boldsymbol{\theta_1}$ and $\boldsymbol{\theta_2}$, which represent the angles between each pendulum and the vertical, respectively.

Step #02: Kinematics – Application of the Constraints

We denote the coordinates of the **CoM for Pendulum-1** and **Pendulum-2** by x_{C_1}, y_{C_1} and x_{C_2}, y_{C_2} respectively. Given the geometry of the pendulum system, it is possible to define the location of the CoM for each pendulum in terms of the generalized coordinates $\boldsymbol{\theta_1}$ and $\boldsymbol{\theta_2}$.

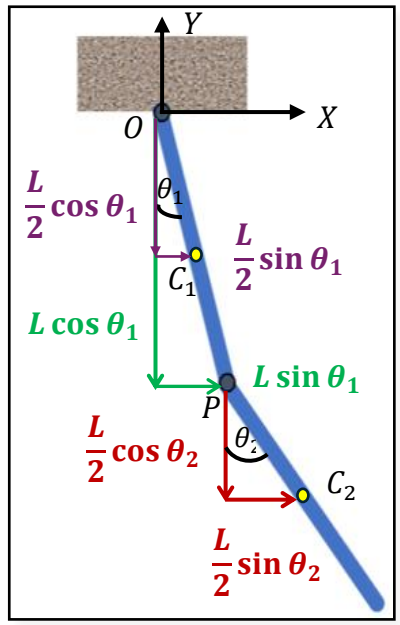
Considering the **origin** of the Cartesian Coordinate System to be at the **pivot point of the 1st pendulum**, the **location of the CoM** for the 2 pendulums can be given as follows:

$$x_{C_1} = \frac{L}{2}\sin\theta_1$$
$$y_{C_1} = -\frac{L}{2}\cos\theta_1$$

Coordinates for the CoM of Pendulum-1 ---------- **(1)**

$$x_{C_2} = L\sin\theta_1 + \frac{L}{2}\sin\theta_2$$
$$y_{C_2} = -L\cos\theta_1 - \frac{L}{2}\cos\theta_2$$

Coordinates for the CoM of Pendulum-2 ---------- **(2)**



Taking time derivatives of the locations, we derive the expressions for the velocity components of the pendulum CoMs, as follows:

$$\dot{x}_{C_1} = \frac{L}{2}\,\dot{\theta}_1\,\cos\theta_1$$
$$\dot{y}_{C_1} = \frac{L}{2}\,\dot{\theta}_1\,\sin\theta_1$$

Velocity of C_1, $V_{C_1} = \sqrt{\dot{x}_{c_1}^2 + \dot{y}_{c_1}^2}$ ---------- **(3)**

$$\dot{x}_{C_2} = L\left(\dot{\theta}_1\,\cos\theta_1 + \frac{1}{2}\,\dot{\theta}_2\,\cos\theta_2\right)$$
$$\dot{y}_{C_2} = L\left(\dot{\theta}_1\,\sin\theta_1 + \frac{1}{2}\,\dot{\theta}_2\,\sin\theta_2\right)$$

Velocity of C_2, $V_{C_2} = \sqrt{\dot{x}_{c_2}^2 + \dot{y}_{c_2}^2}$ ---------- **(4)**

Step #03: Derive Expressions for Kinetic and Potential Energy

To apply Lagrangian Mechanics, we need to determine the kinetic and potential energy of the double pendulum system.

We know that,

The kinetic energy of a rigid body consists of both the translational and rotational elements. Therefore, the kinetic energy of the double compound pendulum can be given as follows:

$$T = \frac{1}{2}m\left(V_{c_1}^2 + V_{c_2}^2\right) + \frac{1}{2}I(\omega_1^2 + \omega_2^2)$$

$$= \frac{1}{2}m\left(\dot{x}_{c_1}^2 + \dot{y}_{c_1}^2 + \dot{x}_{c_2}^2 + \dot{y}_{c_2}^2\right) + \frac{1}{2} * \frac{1}{12}mL^2(\omega_1^2 + \omega_2^2)$$

$$= \frac{1}{2}m\left[\frac{1}{4}\ L^2\dot{\theta}_1^2\{(\cos\theta_1)^2 + (\sin\theta_1)^2\}\right] + \frac{1}{2}m\left[L^2\dot{\theta}_1^2\{(\cos\theta_1)^2 + (\sin\theta_1)^2\} + \frac{1}{4}L^2\dot{\theta}_2^2\{(\cos\theta_2)^2 + (\sin\theta_2)^2\} + L^2\dot{\theta}_1\dot{\theta}_2(\cos\theta_1\cos\theta_2 + \sin\theta_1\sin\theta_2)\right] + \frac{1}{2}mL^2\left(\frac{1}{12}\dot{\theta}_1^2 + \frac{1}{12}\dot{\theta}_2^2\right)$$ **[Putting the values for V_{C_1} and V_{C_2} in terms of θ_1 and θ_2]**

$$= \frac{1}{2}m\left(\frac{1}{4}\ L^2 + \frac{1}{12}\ L^2 + L^2\right)\dot{\theta}_1^2 + \frac{1}{2}m\left(\frac{1}{4}\ L^2 + \frac{1}{12}\ L^2\right)\dot{\theta}_2^2 + \frac{1}{2}mL^2\dot{\theta}_1\dot{\theta}_2\cos(\theta_1 - \theta_2)$$

$$\therefore,\ T = \frac{1}{6}mL^2\left(4\dot{\theta}_1^2 + \dot{\theta}_2^2 + 3\dot{\theta}_1\dot{\theta}_2\cos(\theta_1 - \theta_2)\right) \quad \text{----------} \quad \textbf{(5)}$$

Meanwhile,

The potential energy of the double compound pendulum can be given as follows:

$$V = mg\left(y_{C_1} + y_{C_2}\right)$$

$$= mg\left(-\frac{L}{2}\cos\theta_1 - L\cos\theta_1 - \frac{L}{2}\cos\theta_2\right)$$ **[Putting the values for y_{C_1} & y_{C_2} in terms of θ_1 & θ_2]**

$$\therefore,\ V = -\frac{1}{2}mgL(3\cos\theta_1 + \cos\theta_2) \quad \text{----------} \quad \textbf{(6)}$$

Step #04: <u>Differentiate the Energy Expressions as Required by Lagrange's Equation</u>

To apply the Calculus of Variation based on Hamilton's Principle of Least Action for this pendulum problem, we need a parameter, known as **Lagrangian**, which is defined as follows:

$$L = T - V$$

$$\Rightarrow\ L = \frac{1}{6}mL^2\left(4\dot{\theta}_1^2 + \dot{\theta}_2^2 + 3\dot{\theta}_1\dot{\theta}_2\cos(\theta_1 - \theta_2)\right) + \frac{1}{2}mgL(3\cos\theta_1 + \cos\theta_2) \quad \text{----------} \quad \textbf{(7)}$$

Therefore, the Euler-Lagrange Equation for the pendulum system can be given as follows:

$$\frac{\partial L}{\partial\theta_i} - \frac{d}{dt}\left(\frac{\partial L}{\partial\dot{\theta}_i}\right) = 0$$

$$\Rightarrow\ \frac{d}{dt}\left(\frac{\partial T}{\partial\dot{\theta}_i}\right) - \frac{\partial L}{\partial\theta_i} = 0 \quad \text{----------} \quad \textbf{(8)}$$

We begin by differentiating the Lagrangian with respect to θ_1 and its derivative, respectively and we get,

$$\frac{\partial L}{\partial \theta_1} = \frac{\partial}{\partial \theta_1}\left(\frac{1}{6}mL^2\left(4\dot{\theta}_1^2 + \dot{\theta}_2^2 + 3\dot{\theta}_1\dot{\theta}_2\cos(\theta_1 - \theta_2)\right) + \frac{1}{2}mgL(3\cos\theta_1 + \cos\theta_2\,)\right)$$

$$= \frac{\partial}{\partial \theta_1}\left(\frac{1}{2}mL^2\dot{\theta}_1\dot{\theta}_2\cos(\theta_1 - \theta_2) + \frac{3}{2}mgL\cos\theta_1\right)$$

$$= -\frac{1}{2}mL^2\dot{\theta}_1\dot{\theta}_2\sin(\theta_1 - \theta_2) - \frac{3}{2}mgL\sin\theta_1 \quad \text{---------- } \textbf{(i)}$$

and

$$\frac{\partial T}{\partial \dot{\theta}_1} = \frac{\partial}{\partial \dot{\theta}_1}\left(\frac{1}{6}mL^2\left(4\dot{\theta}_1^2 + 3\dot{\theta}_1\dot{\theta}_2\cos(\theta_1 - \theta_2)\right)\right)$$

$$= \frac{4}{3}mL^2\dot{\theta}_1 + \frac{1}{2}mL^2\dot{\theta}_2\cos(\theta_1 - \theta_2) \quad \text{---------- } \textbf{(ii)}$$

and

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{\theta}_1}\right) = \frac{d}{dt}\left(\frac{4}{3}mL^2\dot{\theta}_1 + \frac{1}{2}mL^2\dot{\theta}_2\cos(\theta_1 - \theta_2)\right)$$

$$= \frac{4}{3}mL^2\ddot{\theta}_1 + \frac{1}{2}mL^2\ddot{\theta}_2\cos(\theta_1 - \theta_2) - \frac{1}{2}mL^2\dot{\theta}_2\left(\dot{\theta}_1 - \dot{\theta}_2\right)\sin(\theta_1 - \theta_2) \quad \text{---------- } \textbf{(iii)}$$

Combining equations (*i*) and (*iii*), we get,

$$\frac{4}{3}mL^2\ddot{\theta}_1 + \frac{1}{2}mL^2\ddot{\theta}_2\cos(\theta_1 - \theta_2) - \cancel{\frac{1}{2}mL^2\dot{\theta}_1\dot{\theta}_2\sin(\theta_1 - \theta_2)} + \frac{1}{2}mL^2\dot{\theta}_2^2\sin(\theta_1 - \theta_2) +$$

$$\cancel{\frac{1}{2}mL^2\dot{\theta}_1\dot{\theta}_2\sin(\theta_1 - \theta_2)} + \frac{3}{2}mgL\sin\theta_1 = 0$$

$$\Rightarrow \frac{4}{3}mL^2\ddot{\theta}_1 + \frac{1}{2}mL^2\ddot{\theta}_2\cos(\theta_1 - \theta_2) + \frac{1}{2}mL^2\dot{\theta}_2^2\sin(\theta_1 - \theta_2) + \frac{3}{2}mgL\sin\theta_1 = 0$$

$$\Rightarrow \boxed{\frac{4}{3}L\ddot{\theta}_1 + \frac{1}{2}L\ddot{\theta}_2\cos(\theta_1 - \theta_2) + \frac{1}{2}L\dot{\theta}_2^2\sin(\theta_1 - \theta_2) + \frac{3}{2}g\sin\theta_1 = 0} \quad \text{---------- } \textbf{(9)}$$

Similarly, by differentiating the Lagrangian with respect to θ_2 and its derivative, respectively and we get,

$$\frac{\partial L}{\partial \theta_2} = \frac{\partial}{\partial \theta_2}\left(\frac{1}{2}mL^2\dot{\theta}_1\dot{\theta}_2\cos(\theta_1 - \theta_2) + \frac{1}{2}mgL\cos\theta_2\right)$$

$$= \frac{1}{2}mL^2\dot{\theta}_1\dot{\theta}_2\sin(\theta_1 - \theta_2) - \frac{1}{2}mgL\sin\theta_2 \quad \text{---------- } \textbf{(iv)}$$

and

$$\frac{\partial T}{\partial \dot{\theta}_2} = \frac{\partial}{\partial \dot{\theta}_2}\left(\frac{1}{6}mL^2\left(\dot{\theta}_2^2 + 3\dot{\theta}_1\dot{\theta}_2\cos(\theta_1 - \theta_2)\right)\right)$$

$$= \frac{1}{3}mL^2\dot{\theta}_2 + \frac{1}{2}mL^2\dot{\theta}_1\cos(\theta_1 - \theta_2) \quad \text{----------} \ (v)$$

and

$$\frac{d}{dt}\left(\frac{\partial T}{\partial \dot{\theta}_2}\right) = \frac{d}{dt}\left(\frac{1}{3}mL^2\dot{\theta}_2 + \frac{1}{2}mL^2\dot{\theta}_1\cos(\theta_1 - \theta_2)\right)$$

$$= \frac{1}{3}mL^2\ddot{\theta}_2 + \frac{1}{2}mL^2\ddot{\theta}_1\cos(\theta_1 - \theta_2) - \frac{1}{2}mL^2\dot{\theta}_1\left(\dot{\theta}_1 - \dot{\theta}_2\right)\sin(\theta_1 - \theta_2) \quad \text{----------} \ (vi)$$

Combining equations (*iv*) and (*vi*), we get,

$$\frac{1}{3}mL^2\ddot{\theta}_2 + \frac{1}{2}mL^2\ddot{\theta}_1\cos(\theta_1 - \theta_2) + \cancel{\frac{1}{2}mL^2\dot{\theta}_1\dot{\theta}_2\sin(\theta_1 - \theta_2)} - \frac{1}{2}mL^2\dot{\theta}_1^2\sin(\theta_1 - \theta_2)$$

$$-\cancel{\frac{1}{2}mL^2\dot{\theta}_1\dot{\theta}_2\sin(\theta_1 - \theta_2)} + \frac{1}{2}mgL\sin\theta_2 = 0$$

$$\Rightarrow \frac{1}{3}mL^2\ddot{\theta}_2 + \frac{1}{2}mL^2\ddot{\theta}_1\cos(\theta_1 - \theta_2) - \frac{1}{2}mL^2\dot{\theta}_1^2\sin(\theta_1 - \theta_2) + \frac{1}{2}mgL\sin\theta_2 = 0$$

$$\Rightarrow \boxed{\frac{1}{3}L\ddot{\theta}_2 + \frac{1}{2}L\ddot{\theta}_1\cos(\theta_1 - \theta_2) - \frac{1}{2}L\dot{\theta}_1^2\sin(\theta_1 - \theta_2) + \frac{1}{2}g\sin\theta_2 = 0} \quad \text{----------} \ (10)$$

Step #05: <u>Deriving the Required Equations of Motion</u>

Therefore, the required equations of motion for a double compound pendulum system are,

$$\boxed{\frac{4}{3}L\ddot{\theta}_1 + \frac{1}{2}L\ddot{\theta}_2\cos(\theta_1 - \theta_2) + \frac{1}{2}L\dot{\theta}_2^2\sin(\theta_1 - \theta_2) + \frac{3}{2}g\sin\theta_1 = 0}$$

$$\boxed{\frac{1}{3}L\ddot{\theta}_2 + \frac{1}{2}L\ddot{\theta}_1\cos(\theta_1 - \theta_2) - \frac{1}{2}L\dot{\theta}_1^2\sin(\theta_1 - \theta_2) + \frac{1}{2}g\sin\theta_2 = 0}$$

Deriving the Equations of Motion using Newtonian Mechanics:

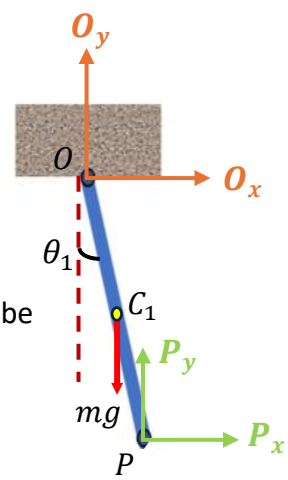
To apply the **Newtonian Mechanics** for the derivation of EoMs of the Double Compound Pendulum System, we use a free-body diagram for each pendulum to consider the forces and moments acting on them individually.

Differentiating the kinematic differential equations (3) and (4), we get the expressions for acceleration as follows:

$$\left.\begin{aligned} \ddot{x}_{c_1} &= \frac{L}{2}\left(\ddot{\theta}_1 \cos\theta_1 - \dot{\theta}_1^2 \sin\theta_1\right) \\ \ddot{y}_{c_1} &= \frac{L}{2}\left(\ddot{\theta}_1 \sin\theta_1 + \dot{\theta}_1^2 \cos\theta_1\right) \end{aligned}\right| \quad \text{---------- (11)}$$

$$\left.\begin{aligned} \ddot{x}_{c_2} &= L\left(\ddot{\theta}_1 \cos\theta_1 - \dot{\theta}_1^2 \sin\theta_1\right) + \frac{L}{2}\left(\ddot{\theta}_2 \cos\theta_2 - \dot{\theta}_2^2 \sin\theta_2\right) \\ \ddot{y}_{c_2} &= L\left(\ddot{\theta}_1 \sin\theta_1 + \dot{\theta}_1^2 \cos\theta_1\right) + \frac{L}{2}\left(\ddot{\theta}_2 \sin\theta_2 + \dot{\theta}_2^2 \cos\theta_2\right) \end{aligned}\right| \quad \text{---------- (12)}$$

Free-Body Diagram of Pendulum-1:



We start with the free-body diagram of the 1st pendulum, where the components of a constraint force acting on the **pivot point O** are denoted by $\boldsymbol{O_x}$ and $\boldsymbol{O_y}$. The reaction force components from Pendulum-2, acting on **point P**, are denoted by $\boldsymbol{P_x}$ and $\boldsymbol{P_y}$.

Therefore, the Equations of Motion of Pendulum-1 **about its CoM at C_1** can be given as follows:

$$\sum F_{x_{c_1}} = m\ddot{x}_{c_1} = O_x + P_x$$

$$\Rightarrow \frac{1}{2} mL\left(\ddot{\theta}_1 \cos\theta_1 - \dot{\theta}_1^2 \sin\theta_1\right) = O_x + P_x \quad \textbf{[Putting the value for } \ddot{x}_{c_1} \textbf{ in terms of } \theta_1\textbf{]}$$

$$\therefore, \boxed{\sum F_{x_{c_1}} = mL\left(\ddot{\theta}_1 \cos\theta_1 - \dot{\theta}_1^2 \sin\theta_1\right) = 2(O_x + P_x)} \quad \text{---------- (13)}$$

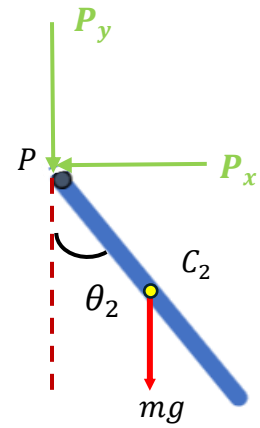
$$\sum F_{y_{c_1}} = m\ddot{y}_{c_1} = O_y + P_y - mg$$

$$\Rightarrow \frac{1}{2} mL\left(\ddot{\theta}_1 \sin\theta_1 + \dot{\theta}_1^2 \cos\theta_1\right) = O_y + P_y - mg \quad \textbf{[Putting the value for } \ddot{y}_{c_1} \textbf{ in terms of } \theta_1\textbf{]}$$

$$\therefore, \boxed{\sum F_{y_{c_1}} = mL\left(\ddot{\theta}_1 \cos\theta_1 - \dot{\theta}_1^2 \sin\theta_1\right) = 2\left(O_y + P_y - mg\right)} \quad \text{---------- (14)}$$

$$\boxed{\sum M^{C_1} = \frac{1}{12} mL^2 \ddot{\theta}_1 = \frac{L}{2}\left(P_x \cos\theta_1 + P_y \sin\theta_1\right) - \frac{L}{2}\left(O_x \cos\theta_1 + O_y \sin\theta_1\right)} \quad \text{---------- (15)}$$

Free-Body Diagram of Pendulum-2:



Moving to the free-body diagram of the 2nd pendulum, we have the reaction force components, $\boldsymbol{P_x}$ and $\boldsymbol{P_y}$, from Pendulum-1, acting on **point P** in the opposite directions.

Therefore, the Equations of Motion of Pendulum-2 **about its CoM at C_2** given as follows:

$$\sum F_{x_{c_2}} = m\ddot{x}_{c_2} = -P_x$$

$$\Rightarrow mL\left(\ddot{\theta}_1 \cos\theta_1 - \dot{\theta}_1^2 \sin\theta_1\right) + \frac{mL}{2}\left(\ddot{\theta}_2 \cos\theta_2 - \dot{\theta}_2^2 \sin\theta_2\right) = -P_x$$

$$\therefore, \boxed{\sum F_{x_{c_2}} = 2mL\left(\ddot{\theta}_1 \cos\theta_1 - \dot{\theta}_1^2 \sin\theta_1\right) + mL\left(\ddot{\theta}_2 \cos\theta_2 - \dot{\theta}_2^2 \sin\theta_2\right) = -2P_x} \text{ -------(16)}$$

$$\sum F_{y_{c_2}} = m\ddot{y}_{c_2} = -P_y - mg$$

$$\Rightarrow mL\left(\ddot{\theta}_1 \sin\theta_1 + \dot{\theta}_1^2 \cos\theta_1\right) + \frac{mL}{2}\left(\ddot{\theta}_2 \sin\theta_2 + \dot{\theta}_2^2 \cos\theta_2\right) = -P_y - mg$$

$$\therefore, \boxed{\sum F_{y_{c_2}} = 2mL\left(\ddot{\theta}_1 \sin\theta_1 + \dot{\theta}_1^2 \cos\theta_1\right) + ml\left(\ddot{\theta}_2 \sin\theta_2 + \dot{\theta}_2^2 \cos\theta_2\right) = 2(-P_y - mg)}$$

-------(17)

$$\boxed{\sum M^{C_2} = \frac{1}{12}mL^2\ddot{\theta}_2 = \frac{L}{2}\left(P_x \cos\theta_2 + P_y \sin\theta_2\right)} \text{ --------(18)}$$

Hence, the required equations of motion for a double compound pendulum system derived using the Newtonian Mechanics are,

$$\boxed{\sum F_{x_{c_1}} = mL\left(\ddot{\theta}_1 \cos\theta_1 - \dot{\theta}_1^2 \sin\theta_1\right) = 2(O_x + P_x)}$$

$$\boxed{\sum F_{y_{c_1}} = mL\left(\ddot{\theta}_1 \cos\theta_1 - \dot{\theta}_1^2 \sin\theta_1\right) = 2(O_y + P_y - mg)}$$

$$\boxed{\sum M^{C_1} = \frac{1}{12}mL^2\ddot{\theta}_1 = \frac{L}{2}\left(P_x \cos\theta_1 + P_y \sin\theta_1\right) - \frac{L}{2}\left(O_x \cos\theta_1 + O_y \sin\theta_1\right)}$$

$$\boxed{\sum F_{x_{c_2}} = 2mL\left(\ddot{\theta}_1 \cos\theta_1 - \dot{\theta}_1^2 \sin\theta_1\right) + mL\left(\ddot{\theta}_2 \cos\theta_2 - \dot{\theta}_2^2 \sin\theta_2\right) = -2P_x}$$

$$\boxed{\sum F_{y_{c_2}} = 2mL\left(\ddot{\theta}_1 \sin\theta_1 + \dot{\theta}_1^2 \cos\theta_1\right) + ml\left(\ddot{\theta}_2 \sin\theta_2 + \dot{\theta}_2^2 \cos\theta_2\right) = 2(-P_y - mg)}$$

$$\boxed{\sum M^{C_2} = \frac{1}{12}mL^2\ddot{\theta}_2 = \frac{L}{2}\left(P_x \cos\theta_2 + P_y \sin\theta_2\right)}$$

Analytical Validation of the EoMs:

To show that both Lagrangian Mechanics and Newtonian Mechanics give the equivalent equations of motion for the double compound pendulum, we will analytically derive the exact Lagrangian EoMs from the Newtonian EoMs by **expressing the constraint and reaction forces** acting on the system **in terms of the generalized coordinates** and **substituting them in the moment equations.**

For our convenience, we consider *m = 1, and L = 1.*

Rewriting Equation (16) and (17), we get,

$$2\left(\ddot{\theta}_1 \cos\theta_1 - \dot{\theta}_1^2 \sin\theta_1\right) + \left(\ddot{\theta}_2 \cos\theta_2 - \dot{\theta}_2^2 \sin\theta_2\right) = -2P_x$$

$$\Rightarrow\ P_x = \left(\dot{\theta}_1^2 \sin\theta_1 - \ddot{\theta}_1 \cos\theta_1\right) - \frac{1}{2}\left(\ddot{\theta}_2 \cos\theta_2 - \dot{\theta}_2^2 \sin\theta_2\right) \quad \text{--------(19)}$$

$$2mL\left(\ddot{\theta}_1 \sin\theta_1 + \dot{\theta}_1^2 \cos\theta_1\right) + ml\left(\ddot{\theta}_2 \sin\theta_2 + \dot{\theta}_2^2 \cos\theta_2\right) = 2(-P_y - mg)$$

$$\Rightarrow\ P_y = -g - \ddot{\theta}_1 \sin\theta_1 - \dot{\theta}_1^2 \cos\theta_1 - \frac{1}{2}\ddot{\theta}_2 \sin\theta_2 - \frac{1}{2}\dot{\theta}_2^2 \cos\theta_2 \quad \text{--------(20)}$$

Putting the values of equations (19) and (20) in equation (18), we get,

$$\frac{1}{6}\ddot{\theta}_2 = \dot{\theta}_1^2 \sin\theta_1 \cos\theta_2 - \frac{1}{2}\ddot{\theta}_2(\cos\theta_2)^2 - \ddot{\theta}_1 \cos\theta_2 \cos\theta_1 + \frac{1}{2}\dot{\theta}_2^2 \sin\theta_2 \cos\theta_2 - g\sin\theta_2 -$$
$$\ddot{\theta}_1 \sin\theta_1 \sin\theta_2 - \dot{\theta}_1^2 \cos\theta_1 \sin\theta_2 - \frac{1}{2}\ddot{\theta}_2(\sin\theta_2)^2 - \frac{1}{2}\dot{\theta}_2^2 \cos\theta_2 \sin\theta_2$$

$$\Rightarrow \frac{1}{6}\ddot{\theta}_2 = \dot{\theta}_1^2(\sin\theta_1 \cos\theta_2 - \cos\theta_1 \sin\theta_2) - \frac{1}{2}\ddot{\theta}_2\{(\cos\theta_2)^2 + (\sin\theta_2)^2\} - g\sin\theta_2$$
$$-\ddot{\theta}_1(\cos\theta_2 \cos\theta_1 + \sin\theta_1 \sin\theta_2)$$

$$\Rightarrow -\frac{2}{3}\ddot{\theta}_2 + \dot{\theta}_1^2 \sin(\theta_1 - \theta_2) - \ddot{\theta}_1 \cos(\theta_1 - \theta_2) - g\sin\theta_2 = 0$$

Dividing the above equation by -2, we get,

$$\Rightarrow \boxed{\frac{1}{3}\ddot{\theta}_2 + \frac{1}{2}\ddot{\theta}_1 \cos(\theta_1 - \theta_2) - \frac{1}{2}\dot{\theta}_1^2 \sin(\theta_1 - \theta_2) + \frac{1}{2}g\sin\theta_2 = 0}$$

Thus, we obtained the equation (10) of Lagrangian Mechanics from equation (18) of Newtonian Mechanics.

Again, rewriting Equation (13) and (14), we get,

$$mL\left(\ddot{\theta}_1 \cos\theta_1 - \dot{\theta}_1^2 \sin\theta_1\right) = 2(O_x + P_x)$$

$$\Rightarrow -O_x = \frac{3}{2}\left(\dot{\theta}_1^2 \sin\theta_1 - \ddot{\theta}_1 \cos\theta_1\right) + \frac{1}{2}\left(-\ddot{\theta}_2 \cos\theta_2 + \dot{\theta}_2^2 \sin\theta_2\right) \quad \text{--------(21)}$$

$$mL\left(\ddot{\theta}_1 \cos\theta_1 - \dot{\theta}_1^2 \sin\theta_1\right) = 2\left(O_y + P_y - mg\right)$$

$$\Rightarrow -O_y = -2g - \frac{3}{2}\left(\ddot{\theta}_1 \sin\theta_1 + \dot{\theta}_1^2 \cos\theta_1\right) - \frac{1}{2}\left(\ddot{\theta}_2 \sin\theta_2 - \dot{\theta}_2^2 \cos\theta_2\right) \quad \text{--------(22)}$$

Putting the values of equations (21) and (22) in equation (15), we get,

$$\frac{1}{6}\ddot{\theta}_1 = P_x \cos\theta_1 + P_y \sin\theta_1 - O_x \cos\theta_1 - O_y \sin\theta_1$$

$$\begin{aligned}\Rightarrow \frac{1}{6}\ddot{\theta}_1 + \ddot{\theta}_1 = &-\frac{1}{2}\ddot{\theta}_2 \cos(\theta_1 - \theta_2) - \frac{1}{2}\dot{\theta}_2^2 \sin(\theta_1 - \theta_2) - g\sin\theta_1 - 2g\sin\theta_1 \\ &-\frac{3}{2}\ddot{\theta}_1\{(\cos\theta_1)^2 + (\sin\theta_1)^2\} - \frac{1}{2}\ddot{\theta}_2(\cos\theta_2\cos\theta_1 + \sin\theta_1\sin\theta_2) \\ &-\frac{1}{2}\dot{\theta}_2^2(\sin\theta_1\cos\theta_2 - \cos\theta_1\sin\theta_2)\end{aligned}$$

$$\Rightarrow \frac{1}{6}\ddot{\theta}_1 + \ddot{\theta}_1 + \frac{3}{2}\ddot{\theta}_1 = -\ddot{\theta}_2 \cos(\theta_1 - \theta_2) - \dot{\theta}_2^2 \sin(\theta_1 - \theta_2) - 3g\sin\theta_1$$

$$\Rightarrow \frac{8}{3}\ddot{\theta}_1 + \ddot{\theta}_2 \cos(\theta_1 - \theta_2) + \dot{\theta}_2^2 \sin(\theta_1 - \theta_2) + 3g\sin\theta_1 = 0$$

Dividing the above equation by 2, we get,

$$\Rightarrow \boxed{\frac{4}{3}\ddot{\theta}_1 + \frac{1}{2}\ddot{\theta}_2 \cos(\theta_1 - \theta_2) + \frac{1}{2}\dot{\theta}_2^2 \sin(\theta_1 - \theta_2) + \frac{3}{2}g\sin\theta_1 = 0}$$

Thus, we obtained equation (9) of Lagrangian Mechanics from equation (15) of Newtonian Mechanics.